Modeling the Adaptive Immune Response to SARS-CoV-2 by Modifying SIM-CoV

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***Abstract*—This project is intended to provide an example and explanation of the immune response to the COVID-19 virus within the human lungs. A simulation of the immune system cells such as epithelial cells, T cells and B cells was generated and investigated for implications about the real-world and the pandemic we are currently living through. The immune system is a very complex system and its components can work together to reduce, and sometimes prevent, incoming viruses. COVID-19 is a very infectious disease, but with the simulation provided by SIM-CoV, we can understand how the human body counteracts the virus. B cells and T cells are the main protectors of the body's immune system. Antibodies and T cells work together to reduce infection and protect from any outside invaders. Vaccines introduce the immune response to a virus to familiarize itself with the body and provide some "memory" for the cells to refer back to. Incoming viruses will be more efficiently reduced and prevented with the "memory" that some cells keep. Variants are also a big topic of discussion and we discovered that variants can be much more contagious because the body's immune system does not recognize the virus and lacks the prevention methods to reduce infection rate. Throughout the paper, there are many more examples of how the immune system operates when confronted with COVID-19.**

***Index Terms*—COVID-19, SIM-CoV, CoranaVirus, Immune Response, Immune System, Lungs, Variants, Vaccination, T Cell, B Cell, Antibodies, Epithelial, Virion, Inflammatory Signals**

I. Introduction

The immune system can be considered a very complex system. In this paper, we will be focusing on how the human body's immune system responds to the COVID-19 virus within the lungs. Lungs are chosen as a central part of the immune system due to the COVID-19 virus being an airborne virus which infects the nasal cavity and then moves to infect the lungs next. Our immune response will activate after the virus has entered our respiratory cells and the ingestion by antigen-presenting cells occurs. Cells named B Cells and T Cells are deployed by the body to combat the incoming virions. A COVID-19 virion is identified as the complete, infective form of the CoronaVirus outside a host cell. A virion has a RNA or DNA and a capsid which impacts the infectivity of the virus. There are many other factors and cells that impact the way our immune system combats viruses. Other factors that influence the immune system include the cells chemokines, antibodies, and lung epithelial cells. Chemokines are small proteins that are secreted by cells that play a role in cell migration through venules from blood into tissue. These proteins influence the immune system by inducing directional movements from the T and B cells which impact the effectiveness of our body's immune response. Epithelial cells are the layers of cells that line the human lungs. These cells act as a kind of barrier which all cells and virions must pass through upon entering the body. Epithelial cells are very important to the immune system because they are the first defense the body has for pathogens such as bacteria and viruses. Lastly, antibodies are a protein that is produced in response to pathogens entering the body. The protein will counteract specific viruses or bacteria. There are many more factors that play a role in the complex system that is a human's immune system, these are just some of the key players that we will focus on within this paper [2].

SIM-CoV, or Spatial Immune Model of CoronaVirus Infection, is a simulation that explores the immune system cells behavior within the human lungs such as virions, epithelial cells, inflammation and immune system cells. Measurements of spatial and temporal dynamics of the virus and immune response will be taken to indicate how exactly our immune system defends the body from the COVID-19 virus. SIM-CoV provides graphical representations of the immune response parameters based on various dynamics that change the behavior of the immune system and its cells. Examples of that include the change of the T cell generation rate, the viral clearance rate, spacing variables and much more. Some key factors that will influence how the simulation works are CD8+ T "killer" cells, B cells, virions, epithelial cells, chemokines, and antibodies. Spatial representation is achieved by three dimensional coordinates (x, y, z) within the simulation that provide a "lung" to occupy with the various types of cells. Time is represented as "time steps" which can be described as one minute per time step. All these dynamics combine together to simulate the complex system that is our own body's immune response within our lungs against COVID-19. The SIM-CoV simulation was provided by Melanie Moses, Judy Cannon, Stephanie Forrest, and Steven Hofmeyr [3].

A. Important Role-Players in the Immune Response

In the most basic sense, an antibody is part of the main defense system of the body to protect against unwanted or dangerous viruses. Antibodies are produced by our body's B cells. B cells have the ability to neutralize viruses so that they are no longer harmful. A B cell does not release the antibodies into the body to fight viruses, instead the antibodies have a binding site that goes into the membrane of the virus to neutralize it. B cells originate from bone marrow. To become active, B cells are matured in the bone marrow. During the maturing process, it is crucial that the receptors develop properly. If a B cell's receptors are not developed properly, then the maturing process of that cell will cease to exist. B cells are able to recognize viruses by the shape of the virus's antigens. Once a virus is recognized a B cell will begin to produce antibodies to fight the antigen. The antibodies that a B cell creates are only present in the body for a couple of months. For instance, the amount of antibodies will increase after a vaccination or an infection, but the peak lifetime of antibodies is limited. B cells can provide long term protection after antibodies wear off. B cells are able to remember antigens and can evolve over time to become better at neutralizing a virus and become more efficient at adapting to various variants of a virus.

There is concern for the waning number of antibodies in our system. Since the number of antibodies reach a peak point after vaccination or infection, it has been recommended that the best protection against severe diseases is achieved through multiple doses of a vaccine. It might seem that vaccines are ineffective because multiple are required for the best protection. It is still possible to get a virus even after a vaccine has been administered. The reason that multiple doses of a vaccine is recommended is because it will increase the count of antibodies in our system and protect us from catching a virus with severe symptoms. Our B cells are always working in the background. Since B cells are able to recognize previous antigens, they can evolve over time to become more efficient at fighting new viruses that enter the body. Similar to many things, without practice our skills for a task will decrease over time. Our B cells are no different. Without new contact of antigens for a long time, the production of antibodies will slow down. Overtime as the production of antibodies slows down, it causes our overall immunity to wane. While the majority of our B cells will decrease, a small handful remain in our body serving as memory cells in case of an invader in the future.

Each year individuals receive a flu shot. It might seem as if we have had one flu shot in the past there should be no reason to get another one. While an instance of this would be favorable, it is not realistic. An antigenic distance is the measurement that is used to determine how similar multiple strains of a virus are. A virus does not always remain the same and is prone to change over a short period of time. While the vaccine that individuals receive each year contains antigens that protect from the flu, the antigens each year are similar but not the same. Equivalent to B cells and their change over time, viruses can also change and evolve over time. A new variant is developed when a mutation occurs on a strand. This mutation could allow for the antigenic structure to change so it will escape from natural or vaccine induced immunity. The effect of waning on antigenically distant variants is similar to B cells. Over time when an antibody mutates, and becomes more distant from the original antibody, the effectiveness of natural or vaccine immunity decreases. Plasma B cells are white blood cells that release antibodies to fight antigens. Plasma cells produce a single type of antibody that has the ability to bind to certain antigens. Each Plasma B cell can produce a couple thousand antibodies per second after an infection is detected.

T cells are a very important part of the human body's auto immune system. They can be described as a type of lymphocyte and play a central role in immunity. T cells are different from B cells due to the fact that T cells only invade and destroy the body's own cells that have been taken over by the virus rather than the virus cells themselves. A lymphocyte is a type of a white blood cell in the immune system. T cells are born from the hematopoietic stem cells found in bone marrow and developed within the thymus gland. This is where the name of T (thymus) in T cells derives from. There are two major, and other minor, subtypes of T cells that have a very important function in controlling and shaping the immune response. The first subtype is the CD8+ "killer" T cell which kills cancer cells or cells that are infected by intracellular pathogens. Intracellular pathogens include cells that contain viruses or bacteria or cells that are damaged in other ways. The second subtype is the CD4+ "helper" T cell which "helps" the activity of other immune cells by releasing cytokines. Cytokines are small protein mediators that alter the behavior of target cells that express receptors for those cytokines[5]. These cells help to polarize the immune response into the appropriate kind depending on the nature of the immunological cells that are present such as virus or bacteria. CD4+ "helper" T cells are very important in adaptive immunity because they help activate B cells and CD8+ "killer" T cells. Both the "killer" and "helper" T cells are very important to the adaptability of the human immune systems and help with the prevention of contracting viruses. Although CD4+ "helper" T cells are very important, SIM-CoV only focuses on the CD8+ "killer" T cells to simulate the impact they have on the COVID-19 virus.

There are many dynamics that can change the way a T cell operates and the effectiveness that cell has on the incoming viruses. T cells have a process to activate and become a strong counteracting force in prevention to foreign viruses. Development of T cells begins inside the bone marrow of the body and then a migration to the thymus occurs. During that migration, each cell can be identified as a subgroup of a T cell, but most will mature into CD4+ or CD8+ T cells. The maturation of a T cell will then begin with the T cell receptor which will allow each T cell to interact with many different types of pathogens and "remember" and recognize those pathogens. If the cell survives this process then it can now become a mature T cell and move onto the next steps of maturation. The next steps include that combination and binding of genes and molecules that develop the T cell to more strongly counteract virions or any other foreign cells. About 98% of T cells die during this process.

The development of T cells provide many dynamics themselves such as the survivability of the cell to maturation, the viruses the cell can recognize, the amount of time it takes to mature, and many other factors. After maturation, T cells can bind to an epithelial cell to improve the "front line" defense that the epithelial cells provide. The binding period and completion of binding to the epithelial cells could influence the effectiveness of the immune response to a virus. If a T cell is able to bind to an epithelial cell at a more efficient rate, then it may be able to effectively prevent more virions entering the body. Arrival time for each cell is a big factor that could influence how efficiently the body responds to viruses as well. Needless to say, time is a big factor that could impact how T cells operate.

Other factors such as the memory of the T cell and how quickly it can combat and dispose of any viruses that it recognizes is also very important. For example, if a person had been infected with COVID-19, then antibodies will be created and the T cell will remember the CoronaVirus. That person's immune system will then be more efficient at preventing the virus from infecting the body again. These cells will eventually die and the body will be more susceptible to contracting the virus again, or if other variants of COVID-19 invade the body. Variants are a big topic of discussion when it comes to this pandemic. The reason why some variants are very infectious is because they are brand new invasive viruses that the immune system has never seen before which reduces that chance of preventing infection.Overall, there are many dynamics that change how T cell's operate and their effectiveness. This also continues to prove that the immune system is a very complex adaptive system.

II. Methods and Results

SIM-CoV gives the user a great deal of freedom and options when simulating the immune response within the human lungs. Research on the immune response mainly took place within the SIM-CoV simulation. We ran multiple tests with varying parameters. Most parameters provided were given with the source code, but we decided to add a couple more parameters to help understand how the simulation works. Before developing the new parameters, we first decided to get a clear understanding of the existing factors of the simulation.

Simulations of the human lung are represented within x, y, and z dimensions. This is where the spatial activity will occur. This also provides a better representation of the physical lung. The Initial infections and virions will be set to kick start the simulation. These initial values will be an integer value that will indicate the number of grid points that will either be a virion or an infection. Time is a very important factor of the SIM-CoV simulation. Many parameters indicate the time periods for the processes of different cell life-cycle stages. For example, the expression, apoptosis T cell binding, and tissue period are all time periods that will indicate how quickly or slowly that stage will take.

Virions are the main aggressor of the simulation and thus, they will have many parameters that will influence the outcome of the simulation. First, the probability of infectivity that the virions produce will help provide an accurate representation of the way COVID-19 could infect the lung. Next, the production rate of virions will indicate how many virions are being produced at a certain time input. A clearance parameter will take a fraction input and express how the virion count decreases. Lastly, a parameter that helps by setting a value to indicate how many virions will diffuse into neighboring cells. This diffusing of neighbor cells means that a virion could diffuse, then turning the grid points around the current diffusing cell will change the surrounding cells to virions. Many of the cells within the immune system are the main influencing factors of the simulation. T cell parameters include the initial delay time that indicates when the cells arrive in the simulation. The vascular tissue and binding period of the T cell are the average minutes before the T cell's death and when it binds to an epithelial cell. The generation rate of T cells will indicate how many T cells are generated per minute in the lung. Finally, the inflammatory signals are the last set of parameters that will influence how the simulation works. The production per minute of inflammatory signals, the decay of the inflammatory signals, and the diffusion of all neighboring grid points are all accounted for. In Fig. 1 (Table 2 in the SIM-CoV paper), you can find a list provided by Dr. Melanie Moses, which indicates all the SIM-CoV parameters and a description of each one.

Parameter	Description	Default
a) Dimensions	Simulation size in x,y, and z dimensions	15000x15000x1
b) Initial Infections	Number of grid points with initial infections	1
c) Initial Virions	Number of virions at initial infection locations	1000
d) Incubation Period	Average minutes until an infected cell starts expressing virions	480
e) Expressing Period	Average minutes after expressing starts until cell death	900
f) Apoptosis Period	Average minutes after apoptosis is induced until cell death	180
g) Infectivity	Probability of one virion infecting one cell per minute	0.001
h) Virion Production	Number of virions produced by expressing cell per minute	1.1
i) Virion Clearance	Fraction by which virion count drops per minute	0.004
j) Virion Diffusion	Fraction of virions that diffuse into all neighboring grid points per minute	0.15
k) Inflammatory Signal Production	Concentration of inflammatory signal produced by expressing cells per minute	1
l) Inflammatory Signal Decay	Fraction by which inflammatory signal concentration drops per minute	0.01
m) Inflammatory Signal Diffusion	Fraction of inflammatory signal that diffuses into all neighboring grid points per minute	1
n) T Cell Production	Number of T cells generated per minute for the whole lung	105000
o) T Cell Initial Delay	Average minutes before T cells start to be produced	10080
p) T Cell Vascular Period	Average minutes before death for a T cell in the vasculature	5760
q) T Cell Tissue Period	Average minutes before death for a T cell after it extravasates	1440
r) T Cell Binding Period	Average minutes T cell is bound to an epithelial cell	10

Fig. 1. Table displaying all main parameters used in the SIM-CoV simulation which is provided by the paper [2]

Although all the existing parameters will efficiently simulate the immune response to COVID-19 within the lungs, there are a couple of parameters that we added that can further influence the results. The first is parameters that allow the user to vary the arrival times of T cells, virions, chemokines, and epithelial cells. With the arrival time being able to be changed, we can simulate very different scenarios in which certain cells may appear later, or certain cells retain memory. We added one other new set of parameters that can influence the effectiveness of both the T cells and the antibodies. This allows the user to simulate scenarios where the body's antibodies and T cells recognize or do not recognize the virus and have a high or low effectiveness towards COVID-19. Influencing parameters throughout the duration of the simulation is another change that our group decided to make. The simulation will consist of no changeable parameters, which could be unlike the real life representation of the immune response. By testing out ever-changing parameters within the simulation, we can simulate many scenarios where the simulation drastically, or slowly changes factors over time. These changes allow for more insight on how the immune system could react in various scenarios that were undiscoverable with the initial set of parameters.

A. Explaining SIM-CoV

SIMCoV is a computational model that simulates cells within the lungs [4]. The simulation focuses on how a virus grows and generates over time. After modeling how a virus generates over time, SIMCoV will display how over many days the virus will deplete and become less infective. Using SIMCoV in UPC++, the particles are modeled on a 3D grid. The Walker paper states that ”... top-down causation – mediated by informational efficacy – plays an important role in dictating the dynamics of living systems,...” In the modeled 3D grid, we can examine the generation of cells over time as a living system [1]. Within the grid, many generations of T cells and B cells inhibit the growth and generations of virions by latching onto the cells and neutralizing them. In Fig. 2, it displays how the Epithelial cells work with T cells to eliminate virions as the number of days increases. During the time period, virions will begin to weaken while the T cells and Epithelial cells grow. Fig. 3 models how virions infect randomly based on close approximation. We see that virions will infect random cells that are close in proximity. Once T cells are introduced then as the days increase the amount of virions will decrease rapidly as T cells grow in population.

B. Baseline Simulation

Knowing how the simulations and the figures work in the previous parts, we then set our sights on doing some more tests that would help simulate different scenarios. The scenario that we simulated was manipulating the effectiveness of the antibodies and T cells within the simulation. With that, the simulation could somewhat show how a body that remembers the virus can prevent it more efficiently. Adding larger values for the effectiveness factors and making the immune system cells arrive at a quicker time helped stimulate an immune system that remembers the virus. With the set up provided, this may be similar to having a vaccine in the body. In the following figures, you will see a list of output values that was generated by the configuration values inserted.

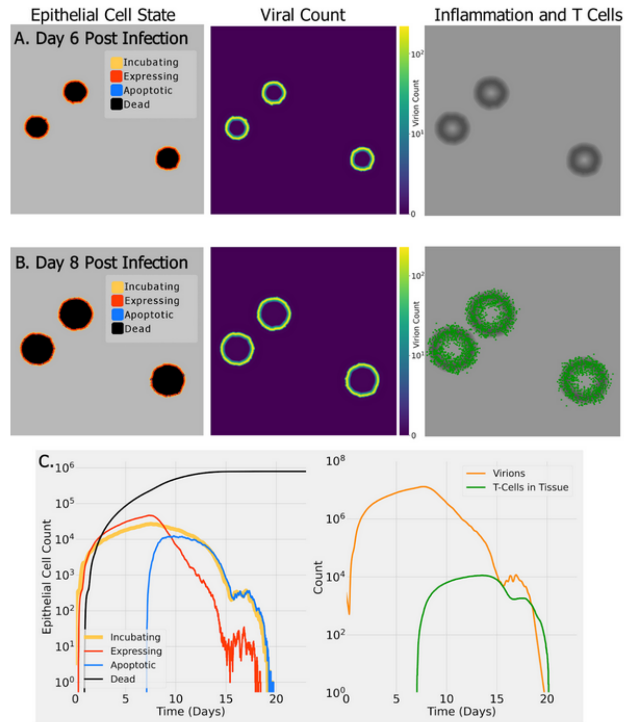

Fig. 2. In this diagram we can see a representation of Epithelial cells, virions, and T cells. The figure illustrates how the strength of the virions will decrease from Day 6 to Day 8 once the Epithelial cells and T cells generate. The graph below illustrates the same concept, by showing how as the time in days increases, the count of the virions drastically decreases on Day 15.

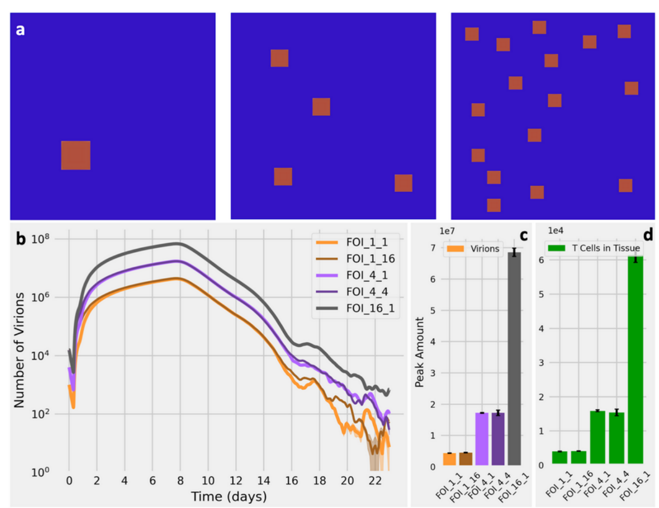

Fig. 3. The blue figure illustrates the effect of Virions and how they populate. Virions will populate randomly and infect those that are close in proximity. In the first panel, there is only one Virion. In the second panel you can see that there are a couple more Virions randomly placed. The random placement is to emphasize the random infection of close proximity. The third pane illustrates the concept even further as there are many more Virions that have populated. The graphs show that as the time(days) increase, the number of Virions will decrease since they become less infective.

# time	incb	expr	apop	dead	tvas	ttis	chem	virs	chempts	%infct
20579	5536	1588	4021	299615	526990490	5028	3.24e-04	5.23e+05	4.32e+05	0.14
20580	5747	1587	4023	299637	527007454	5030	3.24e-04	5.23e+05	4.32e+05	0.14
20581	5739	1540	4031	299661	527024410	5037	3.24e-04	5.23e+05	4.32e+05	0.14
20582	5747	1510	4038	299679	527041356	5031	3.24e-04	5.23e+05	4.32e+05	0.14
20583	5739	1511	4041	299699	527058303	5032	3.24e-04	5.23e+05	4.32e+05	0.14
20584	5738	1514	4033	299724	527075247	5032	3.24e-04	5.23e+05	4.32e+05	0.14
20585	5748	1510	4029	299756	527092188	5035	3.24e-04	5.23e+05	4.32e+05	0.14
20586	5742	1503	4030	299787	527109127	5044	3.24e-04	5.23e+05	4.32e+05	0.14
20587	5745	1508	4039	299804	527126062	5044	3.24e-04	5.23e+05	4.32e+05	0.14
20588	5744	1505	4029	299831	527142997	5049	3.24e-04	5.24e+05	4.32e+05	0.14
20589	5746	1504	4024	299859	527159927	5047	3.24e-04	5.24e+05	4.32e+05	0.14
20590	5744	1505	4029	299879	527176857	5046	3.24e-04	5.24e+05	4.32e+05	0.14
20591	5744	1503	4024	299904	527193786	5045	3.24e-04	5.24e+05	4.33e+05	0.14
20592	5746	1503	4027	299925	527210710	5045	3.24e-04	5.24e+05	4.33e+05	0.14
20593	5746	1502	4034	299943	527227632	5047	3.24e-04	5.24e+05	4.33e+05	0.14

Fig. 4. Results of the initial simulation at the time steps 20579-20593. This simulation sets a good baseline on what the numbers of each variable should be similar to. By comparing the other results and stats to this figure, we can see how the ”variant” and ”vaccine simulation are different and where they are different.

First, we ran a non-modified simulation to set a baseline on what exact differences each of the simulations have. The number shown within the given time gives us an idea of the regular simulation and whether our changes to the code are working or not. In Fig. 4, you can see the baseline values generated and make comparisons with the future figures values.

C. ”Vaccine” Simulation

In Fig. 5, simulation stats can be seen with the effectiveness factor at a higher percentage. The table shows the counts for many stats throughout the simulation but the main stats to focus on are the tvas, ttis, chem and virion stats. The tvas and ttis stats refers to the T cells, the chem stat refers to the chemokines and the virs refers to the virions within the simulation. This gives the effect of higher chemokine counts, T cell stats and a lower virion count. This is expected due to the fact that the immune system was prepared to defend the body against the virus. This does not fully prevent the virus, due to the fact that there are still a great number of virions within the body, but it does prevent it to an extent. Input values that were provided for effectiveness were very small which provides very minimal changes to the numbers compared to other simulation. This small degree of change still indicates that the change for the immune system cells effectiveness can be substantial, even though they are not within our simulation.

# time	incb	expr	apop	dead	tvas	ttis	chem	virs	chempts	%infct
20579	6756	1616	4021	299615	506990490	4938	1.13e-04	7.23e+05	4.31e+05	0.14
20580	6757	1612	4023	299637	507007454	4940	1.13e-04	7.23e+05	4.31e+05	0.14
20581	6756	1611	4031	299661	507024410	4947	1.13e-04	7.23e+05	4.31e+05	0.14
20582	6750	1610	4038	299679	507041356	4947	1.13e-04	7.23e+05	4.31e+05	0.14
20583	6754	1611	4041	299699	507058303	4951	1.13e-04	7.23e+05	4.31e+05	0.14
20584	6754	1614	4033	299724	507075247	4953	1.13e-04	7.23e+05	4.31e+05	0.14
20585	6759	1610	4029	299756	507092188	4955	1.13e-04	7.23e+05	4.31e+05	0.14
20586	6753	1603	4030	299787	507109127	4954	1.13e-04	7.23e+05	4.31e+05	0.14
20587	6744	1608	4039	299804	507126062	4954	1.13e-04	7.23e+05	4.31e+05	0.14
20588	6758	1605	4029	299831	507142997	4959	1.13e-04	7.24e+05	4.31e+05	0.14
20589	6763	1604	4024	299859	507159927	4959	1.13e-04	7.24e+05	4.31e+05	0.14
20590	6762	1605	4029	299879	507176857	4956	1.13e-04	7.24e+05	4.32e+05	0.14
20591	6775	1603	4024	299904	507193786	4956	1.13e-04	7.24e+05	4.32e+05	0.14
20592	6777	1603	4027	299925	507210710	4955	1.13e-04	7.24e+05	4.32e+05	0.14
20593	6772	1602	4034	299943	507227632	4949	1.13e-04	7.24e+05	4.32e+05	0.14

Fig. 5. Results of the ”vaccine” simulation at the time steps 20579-20593. The ”variant” simulation here shows the the numbers are higher than the original baseline simulation number values. This indicates that this introduction of ineffectiveness of antibodies and T cells can impact the way the immune system reacts to a incoming virus. The infectivity rate of the virus is a touch higher on this simulation run as well, which influence the results within the simulation run.

D. ”Variant” Simulation

Trying the opposite by making the effectiveness of the immune system cells less effective was now the plan for the next simulation. Values of the effectiveness were less substantial than the previous run of the simulation, but proved to provide more insight on how tough it is for the immune system to fight against a virus that is very fast acting and new to the body. Once again, the numbers of virions are similar, but there is a significant increase in value within this simulation. Expecting this, we looked into the other values and noticed that there are much less values for the immune system cells counts. This is due to the fact that the effectiveness slider can manipulate how many cells are generated. Overall, this simulation proves that the immune system has a tough time when fending off something that is “new” to the body. This also can indicate that a new variant of COVID-19 can be very infectious due to the ineffectiveness of the immune response, which is why we dubbed Fig. 6, the “variant” simulation.

# time	incb	expr	apop	dead	tvas	ttis	chem	virs	chempts	%infct
20579	4635	544	3366	300477	564240076	5450	6.66e-05	1.58e+05	4.35e+05	0.14
20580	4637	546	3356	300503	564247096	5447	6.66e-05	1.58e+05	4.35e+05	0.14
20581	4634	550	3358	300519	564254112	5448	6.66e-05	1.58e+05	4.35e+05	0.14
20582	4634	552	3353	300542	564261127	5453	6.66e-05	1.58e+05	4.35e+05	0.14
20583	4639	557	3344	300562	564268132	5458	6.66e-05	1.58e+05	4.35e+05	0.14
20584	4643	554	3351	300576	564275144	5463	6.66e-05	1.58e+05	4.35e+05	0.14
20585	4642	552	3360	300594	564282157	5462	6.66e-05	1.58e+05	4.35e+05	0.14
20586	4634	551	3371	300609	564289166	5464	6.66e-05	1.58e+05	4.35e+05	0.14
20587	4625	550	3375	300628	564296179	5459	6.66e-05	1.58e+05	4.35e+05	0.14
20588	4630	551	3370	300648	564303188	5456	6.66e-05	1.57e+05	4.35e+05	0.14
20589	4630	549	3363	300675	564310200	5451	6.66e-05	1.57e+05	4.35e+05	0.14
20590	4631	550	3370	300685	564317211	5451	6.66e-05	1.57e+05	4.35e+05	0.14
20591	4626	551	3369	300705	564324218	5448	6.66e-05	1.57e+05	4.35e+05	0.14
20592	4628	551	3376	300718	564331221	5445	6.66e-05	1.57e+05	4.35e+05	0.14
20593	4634	552	3370	300736	564338225	5444	6.66e-05	1.57e+05	4.35e+05	0.14

Fig. 6. Results of the ”variant” simulation at the time steps 20579-20593. This run simulates with higher effectiveness factors for both antibodies and T cells. Infectivity rate within this simulation is lower than the baseline simulation which indicates that the immune system is more effectively destroying more incoming virus cells. An indication made from this simulation run could relate to how the body reacts in the real-world when vaccinated.

III. Discussion and Conclusions

Our bodies B cells and T cells are some of the main protectors that we have against outside invaders. Both are lymphocytes, white blood cells. Their main job is to activate parts of our immune system to fight against invaders such as viruses and unwanted bacteria. B cells produce antibodies which bind to antigens in an attempt to neutralize. T cells produce cytokines, which work to alert other parts of the immune system to protect from infection. Working together, the innate and adaptive immune system can protect us from many severe cases of viruses. The innate immune system is the body’s first sound of alarm that something is wrong. After the innate immune system sounds the alarm, the adaptive immune system activates. The adaptive immune system holds the T and B cells that use their memory to recognize an invader even if it has been awhile since they have come across the specific antigen. The memory of the cells helps to ensure that even if a virus is caught multiple times, the symptoms don’t become too severe. While T and B cells are able to store memory and recognize previous unwanted antigens, over time their memory will fade if they have not come in contact with a particular invader in a while. The term “while” in our cases is subjective. There is no exact set amount of time that it takes for the antibodies produced by B cells or the cytokines produced by T cells to wane. Waning can vary from one individual to another depending on how the immune system reacts to vaccines.

Knowing that information and the results from our SIM-CoV tests, the COVID-19 virus can be described as a very infectious, fast attacking virus. Many of our tests proved COVID-19 is a virus that can quickly develop inside the human lungs. Our immune system plays a significant role in counteracting the virus. Antibodies and T cells have a significant role in preventing the virus within the lungs. Other cells such as chemokines and epithelial cells also play a great role, but the significance of antibodies and T cells stood to be more apparent to us. With that being said, the simulations also proved that when the body’s cells “remember” the virus, then there is a higher chance of getting lesser symptoms or even prevention of COVID-19. The human immune system is a very complex system, but these cells seem to be working together to help combat any foreign substance that enters the body.

Another observation our group made was that vaccination is a certified way to prevent COVID-19 or lessen the severity of the symptoms if an individual contracts the virus. When antibodies attach themselves to the virus, and allow the other immune system cells, such as T cells, to attack and destroy the substance. Vaccines create a good basis for the immune response to remember COVID-19 due to the fact that the vaccine provides antigens that introduce a non-infectious version of COVID-19 to the body. This can allow the immune response cells to remember and recognize the virus upon coming into contact with it in the future. As stated, these cells do eventually die, which is a reason why booster shots are provided. That way your body can re-learn the virus. Vaccinations provide a great security blanket for the virus, but variants of the COVID-19 virus do exist which implies another set of rules to account for. The reason why variants like the Omicron variant could be more contagious is because this is a new, even faster attacking virus that our body has never seen before. The immune response cells do not know how to prevent the variant because the cells have no memory of it previously, and the rate of infection within the body is much quicker. Our findings found that vaccines and variants are the two major traits in the pandemic that have various factors about each. A more infectious variant would cause more infected people and less security for vaccination. Vaccinations prove to help lessen the severity of symptoms with the variants and are a very important aspect of prevention. The SIM-CoV simulation further does not prove that statement, but by running various tests, it can help provide a great understanding of how it can imply that statement within the human lungs.

Without doubt, the immune system is very complex, complicated, and an art form. An art form meaning that the components of the immune system are all constantly working with one another to transform, notify, and bind[6]. All of these transactions between the various components of the immune system would not be possible unless each transaction was precisely made at the opportune moment to accomplish a task while not interfering with another. While the whole immune system works together, the cells within the immune system also have life cycles of their own. Cells themselves have to work together internally to assure that each of their processes is correct. Correctness of their process matters because if the parts are not put together properly then the cells will not be able to do their job correctly. The behavior of the immune system can be categorized as a very complex and adaptive system.

Our finding most generally focused on the Lungs of the body. The lungs are a very important part of the immune system, and it is the only location of the human immune system that the SIM-CoV simulation focused on. Although the SIM-CoV simulation proved to be vital to understanding the immune system, we can not confidently say that we have a thorough understanding of the immune response to COVID-19 fully. We do have a great deal of understanding on how the immune response reacts to COVID-19 within the lungs, but in other places within the body, such as the nasal, we cannot say the same. Assumptions could be made, but it is never good to assume when dealing with something as complex as the immune system. Variants and vaccinations were brought up in the previous paragraphs, but we did not have a proper opportunity to fully implement these factors within the simulation. We do have results that could implicate these factors are present in the simulation but there is not full proof implementation of vaccines or variants. Some other caveats that occurred include the examination of CD8+ “killer” T cells and not CD4+ “helper” T cells. Both have very strong importance to the immune response and the exploration of both could have benefited our understanding. Although the SIM-CoV simulation is fantastic, it was a challenge to get it running. The UNM CARC machines were a very substantial part of getting the simulation to run, yet we still encountered many issues after getting access. Anything from files not being generated from the simulation to path building errors to the graphs in Paraview not working, we had many issues running the simulation. The reason why we interpreted through the stats file is because we could not find a way to generate the Paraview visualization. After trying it on a PC, Mac, and the CS machines, we bit the bullet and interpreted the way we understood it, which was through the stats files. We think the reason behind our issues when visualizing the simulation was due to the fact that none of our simulations output samples within the samples directory, even after multiple re-builds of SIM-CoV We did what we could and still took away a very important outcome out of this project. The immune system is *very* complex!

IV. Contributions

The topics of this project were learned individually by each team member first. After each team member reviewed the provided resources and content we were able to make modifications to the SIM-CoV. For this project we used Google

Docs to write our findings. Using Google Docs we were able to both edit and annotate the findings. After the text was finalized on Google Docs, we used Overleaf to create the final document. On project 1, we noticed that as we both were editing and typing on Overleaf the live update and syncing of the document with our changes did not fully sync together since new copies of the document were being created too fast. Editing on Google Docs ensured that all of our combined changes were combined. Damian worked to edit the code to vary the time arrival, duration of infection, and the effectiveness. While the code was being edited, Meiling worked to connect the concepts of SIM-CoV to concepts discussed in class. After the modifications to SIM-CoV were complete, both Meiling and Damian interpreted the findings of changing T cell and antibody dynamics.

REFERENCES

[1] S. I. Walker L. Cisneros, P. C. W. Davies, ”Evolutionary Transitions and Top-Down Causation”, Artificial Life 13, Feb, 2012

[2] M. Mitchell, Complexity: A Guided Tour. Oxford: Oxford University Press, 2011

[3] M. Moses, J. Cannon, S. Forrest, S. Hofmeyr, ”Spatially distributed infection increases viral load in a computational model of SARS-CoV-2 lung infection”, 5th Workshop on Viral Dynamics Fred Hutchinson Cancer Research Center, Oct, 2021

[4] SIMCoV, The Regents of the University of California, through Lawrence Berkeley National Laboratory (subject to receipt of any required approvals from the U.S. Dept. of Energy), Arizona State University and University of New Mexico, https://github.com/AdaptiveComputationLab/simcov

[5] A. Iwasaki, ”COVID-19 Immunology 101 for Non-immunologists”, YouTube, Jul, 2020

[6] Flake, G. The Computational Beauty of Nature Chapter 10, 2000